

Ali Barzin

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Professional Summary

Software Engineer specializing in agentic AI and production ML systems, with extensive experience designing, building, and operating cloud-based ML infrastructure. Expert in Python, PyTorch, TensorFlow, LangChain, and LangGraph to develop multi-step reasoning agents and RAG pipelines. Proven track record deploying scalable, monitored, and reliable ML services in cross-functional product teams, with strong focus on responsible AI, evaluation, and optimization of model-backed features.

Technical Skills

- **Programming:** Python, Java, TypeScript, SQL, Bash, JavaScript
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, LangChain, RAG, agentic tool-calling, Transformers, NLP, Machine Learning, LangGraph, Multi-step Reasoning, AI agent frameworks, orchestration
- **Data:** Pandas, NumPy, PySpark, feature engineering, SHAP, Matplotlib
- **Infrastructure:** Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD, REST/OpenAPI, AWS, Cloud-based ML Infrastructure, distributed ML systems

Professional Experience

SecureAuth Corporation | *Software Engineer - Agentic AI & ML-Driven Security Platform* Jan 2022 - Apr 2025

- Led reliability automation for ML-driven risk engines using TypeScript and feature-engineering techniques, integrating location, browser, and time signals; partnered with product, engineering, and data teams to define compliance-ready ML requirements, leveraging multi-step reasoning to improve detection accuracy and reduce false-positive alerts
- CI/CD (GitHub Actions, Jenkins); Docker/Kubernetes for reproducible model-backed deployments on cloud-based ML infrastructure, operating 61+ ActiveAdmin panels and 28+ background jobs across SecureAuth risk and authentication services
- Python platform code for monitoring, evaluation, and optimization of model behavior, configuration lifecycle, and pre-release gates

Acceptto Corporation | *Software Engineer - Agentic AI & ML Infrastructure* Sep 2016 - Jan 2022

- Developed systems for unsupervised anomaly detection, pattern recognition, and continuous authentication fingerprint scoring on a high-concurrency platform using Java, JavaScript (283+ service modules), and Kubernetes, which improved detection of abnormal login attempts and reduced risky session persistence
- Java Android SDK modules and REST APIs for behavioral authentication; Python services to exercise agentic ML workflows end-to-end with AI agent frameworks such as LangChain and LangGraph for orchestration of multi-step reasoning agents
- Automated release checks in CI for mission-critical ML surfaces as part of cloud-based ML infrastructure, covering 199+ domain models and 388+ database migrations to keep authentication, risk, and policy services deployable with confidence

Entrepreneurial Leadership | *Strategic Systems Consultant (Data-Driven Operations)* Jan 2005 - Jan 2014

- Developed and deployed data-driven forecasting and risk-analysis strategies for finance and operations using Python, statistical regression techniques, and spreadsheet-based models; built predictive forecasting and asset-risk models that improved risk-assessment accuracy and supported more informed decision-making

Selected Projects

Rai | [Rai](#) May 2026 - Present

- Design and build an agentic AI advisor in Python using AI agent frameworks such as LangChain and LangGraph and RAG for multi-step reasoning; implement 12 API routers, 7 specialist agents, and ~23 concrete tools across 7 tool domains with autonomous tool orchestration, RBAC, confirmation prompts, and step-up MFA; add monitoring, evaluation, and cost/latency guardrails plus OWASP-aligned alignment and CI-backed OpenAPI contract checks for reliable production behavior

Asar | [Asar](#) Nov 2025 - Present

- Integrate Rai into admin UX and mobile (Entros) via a Python-based gateway, exposing agentic AI capabilities as production services backed by 13 microservices, 63 Go handler modules (36 identity, 27 dashboard), and 6 anomaly/risk analyzers; wire in Trust services for authentication, MFA, audit logging, 6 social IdP integrations, and dynamic risk assessment; containerize with Docker and orchestrate via Kubernetes as a distributed ML system with 128+ dashboard REST routes, 33 React admin UI pages, and a TypeScript browser SDK on scalable, cloud-based ML infrastructure with monitoring and reliability guardrails

Movie Agent Service | [Movie Agent Service](#)

- Built an agentic AI movie recommendation service using AI agent frameworks such as LangChain to compare tool-calling orchestration vs. ReAct-style multi-step reasoning; implemented schema-driven tools and service-layer state/validation for reliable, auditable agent behavior; deployed a LangChain/FAISS/Flask RAG stack with live Hugging Face demo to showcase production-ready API integration and lower hallucination rates

Fraud Shield AI | [Fraud Shield AI](#)

- Designed a PySpark + PyTorch fraud-detection pipeline as a distributed ML system for high-volume transactional data; implemented feature engineering, model training, and evaluation loops with monitoring to optimize precision/recall; prepared cloud-based deployment hooks for scalable, production fraud-scoring services

Churn Intelligence | [Churn Intelligence](#)

- Developed a cloud-ready churn prediction service as a distributed ML system using XGBoost ensembles and SHAP for model explainability; implemented training, evaluation, and monitoring workflows to optimize ROC-AUC (84.77%) and enable responsible AI in customer-retention decisions; exposed predictions and explanations via APIs suitable for scalable, production ML infrastructure

Education & Training

Interview Kickstart

AI/ML SWE

- **Achievements:** 800/800 ML & System Design

Industrial Management Institute (IMI)

Executive MBA (GPA: 3.7)

Allameh Tabataba'i

B.A., Economics (Econometrics)

Portland CC

Web Certificate (GPA: 3.9)